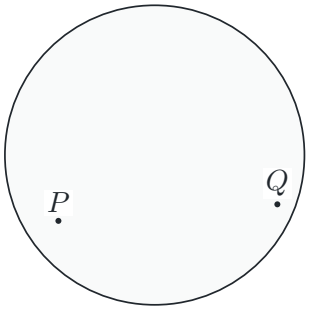
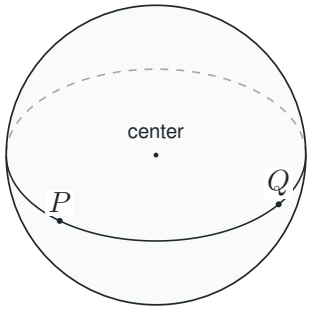
Keep the ball on its support. Keep the string against its surface and its endpoints on the same two markers. Dashed arcs lie on the back of the ball.

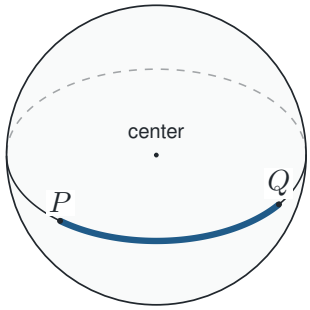
A *great circle* is the circle made by a flat cut through the ball's center. A shortest surface route between two points runs along a great circle through them.



Mark P and Q



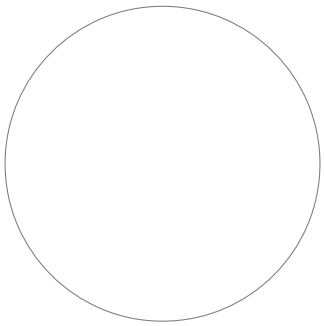
Their great circle



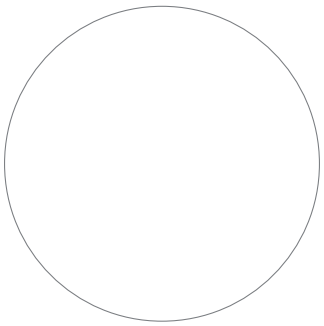
Shorter arc: P to Q

Problem 1: Choose a close pair of markers, a far-apart pair that is not opposite, and a pair exactly opposite each other on your ball. Find the shortest surface routes for each pair. Which pairs let you find two different routes of the same shortest length? Draw your routes.

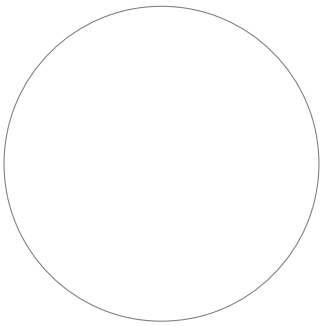
Close pair



Far-apart pair

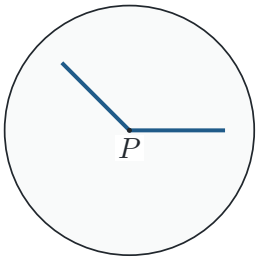


Opposite pair

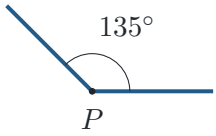


Join three different markers by their shortest surface routes. Do not use opposite markers or put all three on one great circle. Use the smaller region inside the three sides.

An angle on the ball is the angle between the directions of the two sides at their meeting point. Use the inside corner.



Two sides at P

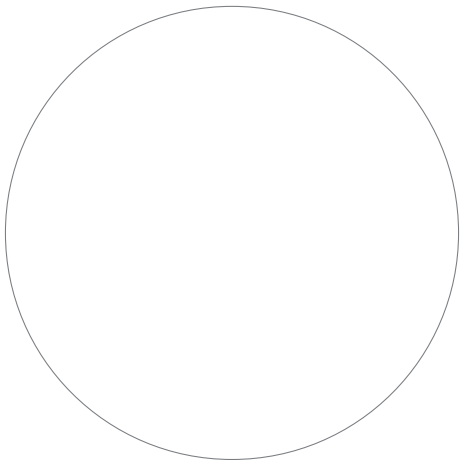


Directions close to P

$P:\ 135°$

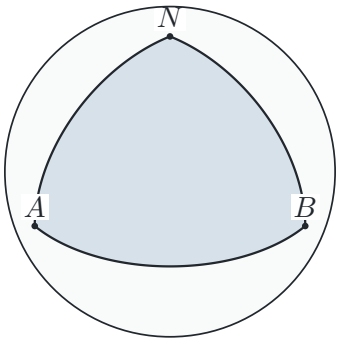
Angle record

Problem 2: Make a triangle on your ball with as many right-angle corners as you can. Check its corners with the paper right angle. Can all three corners be right angles? Draw your best triangle and mark its corners.



The equator is a great circle. N is the pole above it. A meridian is a great-circle semicircle from N to the opposite pole.

Problem 3: Keep N fixed. Move A and B on the equator, using its shorter arc as the third side. Make triangles whose corner at N is smaller than a right angle, equal to a right angle, and larger than a right angle. Draw each triangle and mark all its corners. Which changes make the triangle cover more of the ball?



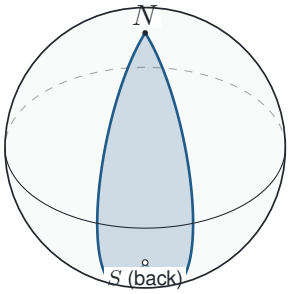
N corner smaller than a right angle

N corner equal to a right angle

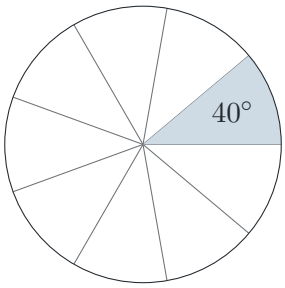
N corner larger than a right angle

Extra workspace

A *lune* is the strip between two great-circle semicircles from N to its opposite pole S. The strip's angle is the gap between them at N. Fractions refer to the ball's surface.



$40°$ lune



View down toward N

9 equal lunes fill the ball

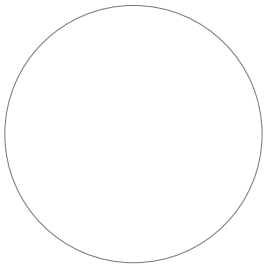
$360 \div 40 = 9$

One lune: $\frac{1}{9}$ of the surface

Area record

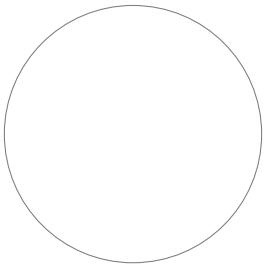
Problem 4: Use the marked meridian pairs with gaps $45°$, $60°$, and $120°$. For each pair, make a triangle with N and the two equator points. What fraction of the ball's surface does each triangle cover?

$45°$ gap



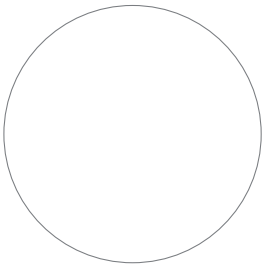
Fraction: _____

$60°$ gap



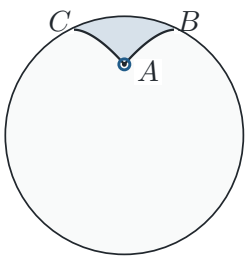
Fraction: _____

$120°$ gap

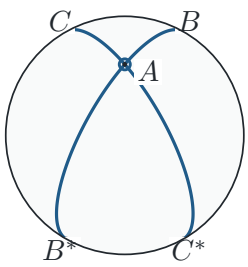


Fraction: _____

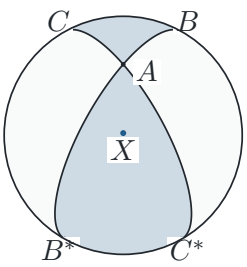
At a triangle corner, extend its two meeting sides to full great circles. Choose the lune containing the triangle and the opposite lune. Together they are a *lune pair*. A starred point is exactly opposite its matching letter.



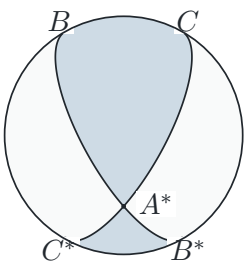
Corner A



Circles AB and AC



Pair at A: front

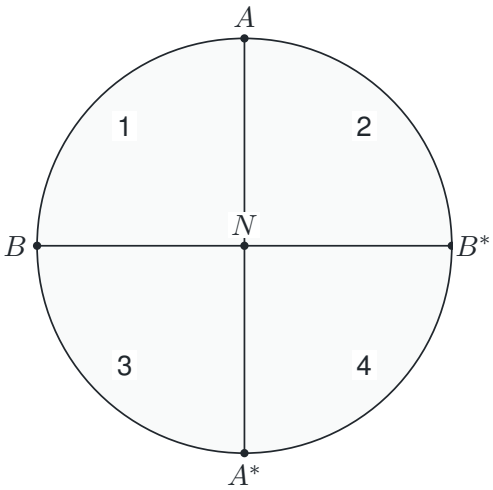


Same pair: back

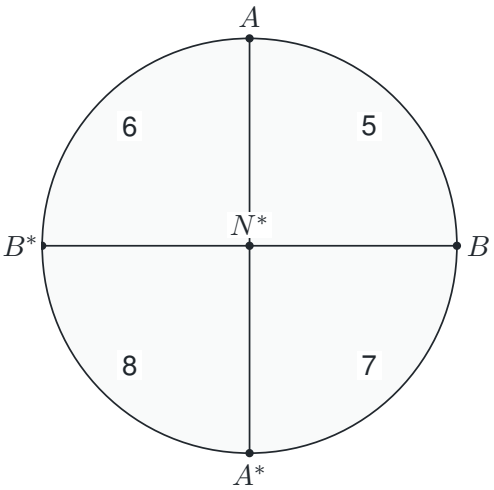
Marked pair	Patch	Count
A	X	1

Each pair covers a patch once or not at all. Front and back show the same ball; these views omit hidden arcs.

Problem 5: Use a triangle with three right angles on your ball. Extend its sides to full great circles. For each of its three corners, mark the lune pair. How many of the three pairs cover each of the eight regions? Use your counts to explain what fraction of the ball the triangle covers.



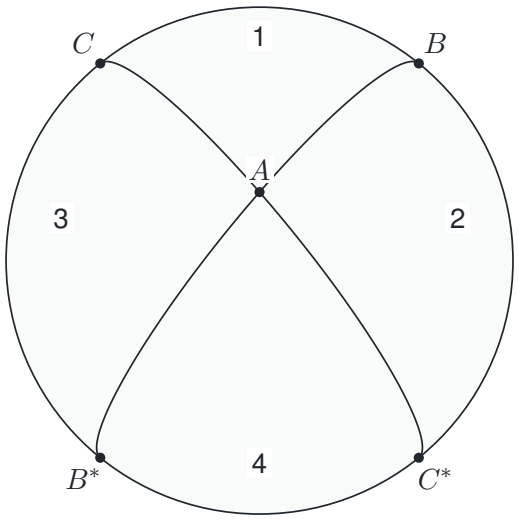
Front



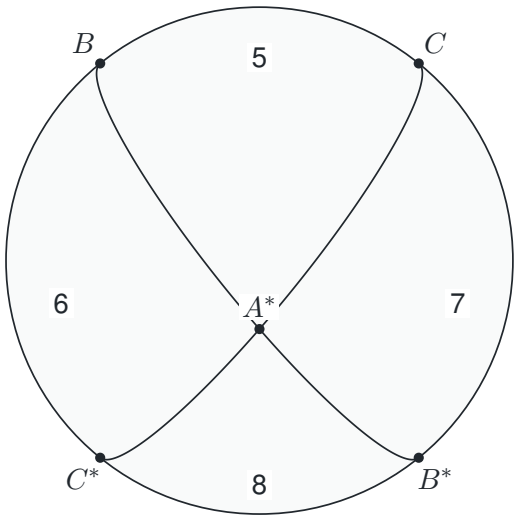
Back

Region	1	2	3	4	5	6	7	8
Pairs covering it	____	____	____	____	____	____	____	____

Problem 6: The triangle ABC has three $80°$ surface corners. Use these views to mark which pairs, at A, B, and C, cover each region. Do the unequal regions give the same counts as Problem 5? Use the pair areas and your counts to find the triangle's fraction of the ball.



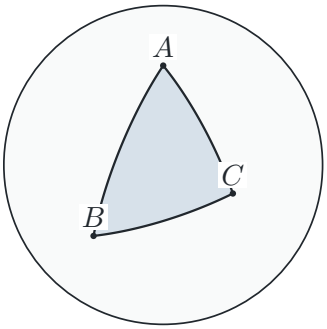
Front



Back

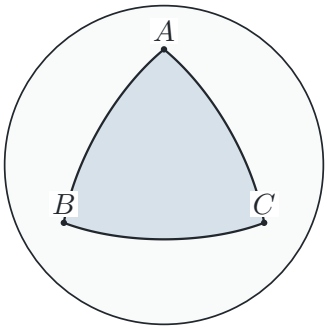
Region	1	2	3	4	5	6	7	8
Pair letters	____	____	____	____	____	____	____	____
Count	____	____	____	____	____	____	____	____

Problem 7: Find a rule for a triangle's fraction of the ball using the sum of its three surface corners. Use it for these three triangles. Explain why your rule works for every triangle under our rules.



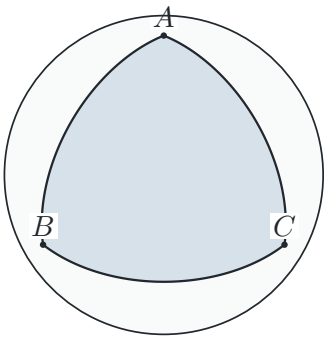
$A = 60°,\ B = 60°,\ C = 90°$

Fraction: _____



$A = B = C = 80°$

Fraction: _____



$A = B = C = 100°$

Fraction: _____